A Proof of the Poset Pattern Avoidance Conjecture

$\mathrm{Av}_P(\{1\}\{1,2\}\{2\}) \le \mathrm{Av}_P(\{1\}\{2\}\{1,2\})$

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1 Introduction and Setup

A permutation on an n-element poset P is a list $\sigma = (\sigma_1, \ldots, \sigma_n)$ of the elements of P. We say σ contains a permutation pattern $\pi = (\pi_1, \ldots, \pi_k)$ on a k-element poset Q if there exists a subsequence $\sigma_{i_1}, \ldots, \sigma_{i_k}$ order-isomorphic to π. Let $\mathrm{Av}_P(\pi)$ denote the number of permutations on P that avoid π.

In arXiv:1208.5718, Morgan Weiler and Sam Hopkins showed that $\mathrm{Av}_P(132) \le \mathrm{Av}_P(123)$ for all finite posets P. Here we consider their subsequent conjecture: whether $\mathrm{Av}_P(\{1\}\{1,2\}\{2\}) \le \mathrm{Av}_P(\{1\}\{2\}\{1,2\})$ for all finite posets P.

Write $a \parallel b$ for incomparable elements in P.

- An **A-occurrence** is a triple $i < j < k$ with $w_i, w_k < w_j$ and $w_i \parallel w_k$ (corresponding to pattern $\{1\}\{1,2\}\{2\}$).
- A **B-occurrence** is a triple $i < j < k$ with $w_i, w_j < w_k$ and $w_i \parallel w_j$ (corresponding to pattern $\{1\}\{2\}\{1,2\}$).

Either kind *ends* at k and has *middle* j. Note that B is symmetric in its first two entries.

2 The Swap Map and Termination

If w contains B, let k be the least end position of a B-occurrence, and let j be the least middle among B-occurrences ending at k. Let $s(w)$ be w with positions j and k swapped. Put $x = w_j$ and $y = w_k$, and fix a witness $i < j$ with $u = w_i$, so $u, x < y$ and $u \parallel x$. In $s(w)$, positions i, j, k hold u, y, x, forming an A-occurrence.

For an A-avoider w^0, define $w^{t+1} = s(w^t)$ while w^t contains B, and let $\Phi(w^0)$ be the terminal permutation. This process terminates: for a linear extension $\lambda\colon P \to [n]$ and potential $H(w) = \sum_r r\,\lambda(w_r)$, each swap changes H by $-(k-j)\big(\lambda(y) - \lambda(x)\big) < 0$. Thus $\Phi(w^0)$ avoids B.

3 Structural Claims

Let $w' = s(w)$, with i, j, k, u, x, y defined as above.

Claim 1. *w' has no B-occurrence ending before k.*

Proof. One ending before j is unchanged from w. One ending at j consists of two incomparable elements before j below y, giving a B-occurrence in w ending at k with middle $< j$. One ending at $h \in (j, k)$ must use position j, so $y < w_h$, yielding a B-occurrence (i, j, h) in w ending before k. All cases contradict minimality. □

Claim 2. *If w has no A-occurrence ending after k, then neither does w'.*

Proof. Let (p, q, r) be an A-occurrence in w' with $r > k$. Case analysis on $\{p, q\}$ intersecting $\{j, k\}$ shows that any such occurrence forces a corresponding A- or B-occurrence in w violating the minimality of k or the hypotheses on w. □

Claim 3. *Under the hypothesis of Claim 2, the greatest end position of an A-occurrence in w' is k, and among those ending at k, the greatest middle is j.*

Proof. (i, j, k) is an A-occurrence in w', and Claim 2 rules out later ends. A putative A-occurrence (p, h, k) with $h > j$ implies a B-occurrence in w ending at $h < k$, contradicting the minimality of k. □

4 Injectivity and Conclusion

Theorem 1. *For any finite poset P, we have* $\mathrm{Av}_P(\{1\}\{1,2\}\{2\}) \le \mathrm{Av}_P(\{1\}\{2\}\{1,2\})$.

Proof. For w containing A, let $r(w)$ swap the middle and end of the A-occurrence with greatest end, then greatest middle. By Claims 1–3, the forward swap chain preserves the hypothesis of Claim 2, and the reverse map r uniquely reconstructs w^0 from $\Phi(w^0)$. Thus Φ is an explicit injection from $\mathrm{Av}_P(\{1\}\{1,2\}\{2\})$ to $\mathrm{Av}_P(\{1\}\{2\}\{1,2\})$. □